SYNOPSIS
ON

Brain Tumor Detection Using Microstrip Patch Antenna

SUBMITTED BY:

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Abstract

Brain tumours are serious neurological disorders that require early detection for effective treatment. Conventional imaging techniques such as MRI and CT scans are accurate but expensive and not easily accessible in rural areas. This project proposes a non-invasive microwave-based method for brain tumour detection using a microstrip patch antenna operating at 2.4 GHz.

The system works on the principle of dielectric contrast between normal brain tissue and tumour tissue. Since tumour tissues have higher water content, they exhibit higher permittivity, which causes measurable changes in the reflection coefficient (S11) of the antenna. A gel-based head phantom model is prepared to simulate brain tissue, and a tumor model is embedded inside it.

The antenna is connected to a Vector Network Analyzer (VNA) to measure reflection parameters under normal and tumor conditions. The measured data is analyzed using MATLAB to observe resonant frequency shifts. The results demonstrate that the presence of a tumor causes a noticeable frequency deviation, validating the feasibility of the proposed detection method.

Introduction

Brain tumour detection is traditionally performed using imaging techniques such as MRI and CT scans. While effective, these systems are expensive and not easily accessible in rural or low-resource areas. Microwave-based detection techniques offer a low-cost, non-ionizing, and portable alternative.

Microwave imaging works on the principle that biological tissues exhibit different dielectric properties. Tumor tissues typically contain higher water content, leading to higher permittivity and conductivity compared to normal brain tissue. When microwave signals interact with these tissues, variations in reflected signals can be observed.

This research focuses on detecting these variations using a microstrip patch antenna and analyzing reflection parameters.

BLOCK Diagram

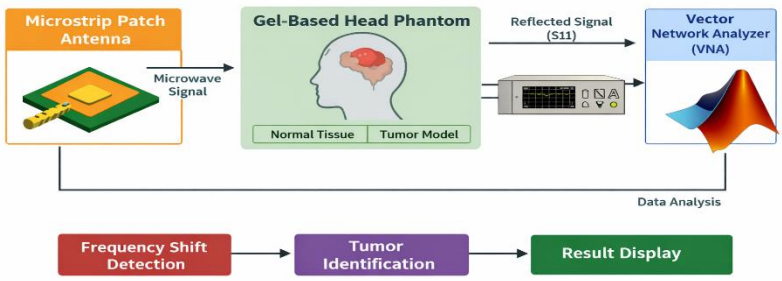


Fig no.1

Working & Principles

Principle of Operation

The proposed brain tumour detection system is based on the principle of **microwave dielectric contrast imaging**. Biological tissues possess specific electrical properties such as permittivity (ε), conductivity (σ), and dielectric constant. These properties determine how electromagnetic waves interact with the tissue.

Tumor tissues generally contain higher water content compared to normal brain tissues. Due to this increased water content, tumor tissues exhibit higher permittivity and conductivity. When microwave signals are transmitted toward the brain, the difference in dielectric properties between healthy and tumor tissues causes variation in the reflected electromagnetic waves.

The reflection coefficient, commonly represented as **S11**, indicates how much of the transmitted signal is reflected back to the antenna. Any change in internal tissue composition results in a measurable shift in the resonant frequency and reflection magnitude. By analyzing these variations, the presence of a tumor can be detected.

Thus, the detection mechanism is based on identifying frequency shifts and amplitude changes in the reflection parameter caused by dielectric contrast inside the brain model.

Working of the System

The system consists of a microstrip patch antenna operating at 2.4 GHz, a gel-based head phantom model, and a Vector Network Analyzer (VNA).

1. The microstrip patch antenna is designed and fabricated on an FR4 substrate to operate at the desired frequency.
2. The antenna is connected to a Vector Network Analyzer to measure the reflection coefficient (S11).
3. Initially, the antenna is tested in free space to obtain baseline resonance characteristics.
4. A gel-based head phantom model is prepared to simulate the electrical properties of normal brain tissue.
5. The antenna is placed at a fixed distance from the phantom, and the reflection coefficient is recorded.
6. A tumor model with higher water concentration is embedded inside the phantom to simulate abnormal tissue.
7. The S11 parameter is measured again under the tumor condition.
8. The measured data is exported and analyzed using MATLAB to observe shifts in resonant frequency and changes in reflection magnitude.

Applications

1. Early-stage tumor screening.
2. Rural healthcare diagnostic systems.
3. Portable medical devices.
4. Research in biomedical microwave imaging.

Future Scope

1. The system can be upgraded using an antenna array to improve detection accuracy and sensitivity.
2. Advanced signal processing techniques can be applied to develop a complete microwave imaging system for better tumor localization.
3. Machine Learning algorithms can be integrated for automatic classification of normal and tumor conditions.
4. Future research can focus on estimating tumor size and position more precisely.
5. The proposed model can be developed into a portable and low-cost diagnostic device for rural healthcare applications.

Conclusion

This project presents a microwave-based non-invasive technique for detecting brain tumors using a microstrip patch antenna operating at 2.4 GHz. The detection mechanism is based on the dielectric contrast between normal brain tissue and tumor tissue. Since tumor tissues have higher water content, they exhibit different electrical properties, which cause measurable variations in the reflection coefficient (S11).

By measuring and comparing the resonant frequency under normal and tumor conditions using a Vector Network Analyzer (VNA), a noticeable frequency shift is observed in the presence of a tumor. The recorded data is further analyzed using MATLAB to validate the variation in reflection characteristics.

The proposed system offers a safe, low-cost, and portable alternative to conventional imaging techniques. Although it is developed at a prototype level using a head phantom model, the results confirm the feasibility of microwave-based tumor detection and open opportunities for further research and improvement.

References

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